

llmXive Automated Review of WeaveBench: A Long-Horizon, Real-World Benchmark for Computer-Use Agents with Hybrid Interfaces

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper presents strong quantitative claims, but several narrative statements overgeneralize specific model results or omit necessary data points for verification.

In Section 4.2, the text asserts that “the hybrid gain is +31.6pp” as a general property of WeaveBench. This figure is derived exclusively from the Claude Opus 4.7 results (35.1% Hybrid vs 3.5% CLI). However, Table 4.1 shows GPT-5.5 achieves a gain of 30.7pp (33.3% vs 2.6%). Presenting the single best model’s gain as the benchmark’s general characteristic risks misleading readers about the average difficulty or the consistency of the hybrid requirement across different architectures. The authors should clarify that this figure represents the gain for the strongest model or provide an average across the tested backbones.

In Section 4.3, the claim that the trajectory-aware judge removes “between 10.3 and 20.2 PassRate points across the four GPT backbones” is difficult to verify from the provided text. While the 20.2pp drop for GPT-5.5 (from 53.5% to 33.3%) is explicitly supported by Table 4.2 and Figure 4.2, the text does not list the corresponding “outcome-only” inflated scores for GPT-5.4, 5.3, and 5.2. Without these specific numbers, the lower bound of 10.3pp remains an unsupported assertion in the narrative. The authors should include the specific inflated scores for the other backbones in the text or a referenced table to substantiate the claimed range.

Finally, in Appendix C, the statement that CLI and Vision channels are “on par” (79.1% vs 77.3%) and that CLI takes “half the steps” (14.3 vs 29.0) lacks the precision expected in a benchmark paper. While 1.8% is a small difference, “on par” is a qualitative judgment that should be qualified (e.g., “statistically comparable”) or supported by a significance test. Similarly, 14.3 is approximately 49% of 29.0; phrasing this as “half” is a slight exaggeration. Using “approximately half” and “statistically comparable” would align the prose more rigorously with the data.

Action items.

- **[writing]** Section 4.2 claims a ‘+31.6pp’ hybrid gain for WeaveBench generally, but this figure is specific to Claude Opus 4.7 (35.1% vs 3.5%). GPT-5.5 shows ~30.7pp. Clarify if this is a model-specific or average gain to avoid overgeneralization.
- **[writing]** Section 4.3 states the judge removes ‘10.3 to 20.2’ points across four GPT backbones. While 20.2pp for GPT-5.5 is shown, the text lacks the specific inflated scores for the other three backbones needed to verify the 10.3pp lower bound. List these scores.
- **[writing]** Appendix C claims CLI and Vision are ‘on par’ (79.1% vs 77.3%) and CLI takes ‘half the steps’ (14.3 vs 29.0). Qualify ‘on par’ as ‘statistically comparable’ and ‘half’ as ‘approximately half’ to maintain numerical precision.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 effectively illustrates the three hybrid workflows with clear step-by-step visualizations, but the small text in the summary boxes and some blurry terminal outputs reduce readability and precision.

Figure 2

The figure provides a clear visual overview of the pipeline stages, but the caption contains a minor typo ('\$3') and redundantly repeats text found in the figure itself.

Figure 3

The figure effectively visualizes the dataset taxonomy and distribution metrics, but the caption contains a typo ('\$') and the y-axis label in panel (b) is ambiguous regarding what is being counted.

Figure 4

The figure effectively visualizes failure anatomy, but the caption for panel (a) is misleading as it claims to show a distribution across three backbones while the chart only displays an aggregated total. Additionally, the legend mapping for the sunburst chart is implicit rather than explicit.

Figure 5

Figure 5 effectively visualizes the long-horizon and cross-application nature of the tasks with clear statistical annotations, though the y-axis labels suffer from truncation and low resolution, and the distribution in panel (b) warrants a closer look to ensure the 'at least one switch' claim aligns with the visual data in the first bin.

Figure 6

The figure effectively visualizes P2 and P3 metrics by domain, but the horizontal dotted lines representing the stated thresholds (20 calls and 3 apps) are visually misaligned with the y-axis values described in the caption.

Figure 7

The figure is visually clear and the data is legible, but the caption contains a grammatical error with a missing noun. Additionally, there is a discrepancy between the legend and the chart data, as a dark blue segment in the ‘GAM’ and ‘SPA’ bars is not defined in the legend.

Action items.

- **[writing]** Figure 1: The right-side summary boxes contain text that is significantly smaller and lower contrast than the main workflow labels, making the ‘Only CLI’ vs ‘Only GUI’ distinctions difficult to read.
- **[science]** Figure 1: The ‘DAV’ workflow Step 1 shows a Jaeger trace with a red leaf span, but the specific ‘payments.fx-rate.http’ span is not clearly highlighted or labeled in the screenshot to match the text description.
- **[writing]** Figure 1: The ‘GAME’ workflow Step 2 terminal output lists ‘3 issues’ but the text is blurry and the specific file paths (e.g., ‘scenes/main.tscn’) are hard to verify against the step description.
- **[writing]** Figure 2: The caption contains a typo ‘stress-tested by \$\$3 pilot agents’ with a stray dollar sign.
- **[writing]** Figure 2: The caption text ‘Task: 114 tasks across 8 domains’ is repeated verbatim in the figure’s ‘Pilot Runs’ section, creating redundancy.
- **[writing]** Figure 3 caption: contains a typo ‘GUI \$\$ CLI’ instead of ‘GUI / CLI’ or ‘GUI and CLI’.
- **[science]** Figure 3(b): the y-axis label ‘Switches’ is ambiguous; it should explicitly state ‘Number of tasks’ to match the x-axis label and clarify that bars represent task counts per switch range.
- **[science]** Figure 4: The caption describes panel (a) as showing ‘Overall error distribution across the three frontier backbones’, but the rendered sunburst chart displays a single aggregated distribution (n=1,735) without distinguishing the three backbones (Opus 4.7, GPT-5.5, GPT-5.4) which are only shown in panel (b).
- **[writing]** Figure 4: The legend at the top of panel (a) uses color blocks for categories (E1 Reasoning, E2 Tool/Exec, etc.) that are not explicitly labeled in the chart’s inner ring, requiring the reader to infer the mapping between the legend colors and the inner ring segments.
- **[writing]** Figure 5: The y-axis label ‘# tasks’ is truncated to ‘# tasks’ in panels (a) and (b), and the decimal point in the y-axis tick labels (e.g., ‘10.0’, ‘12.5’) is barely legible due to low resolution.
- **[science]** Figure 5: Panel (b) displays a histogram of ‘GUI <-> CLI channel switches per task’ with a median of 16, yet the x-axis extends to 70 with sparse bars; the caption claims ‘every task has at least one switch’ but the first bin (0-5) appears to contain significant counts, creating a potential visual contradiction regarding the minimum value.

- **[science]** Figure 6: The caption states dotted lines mark the P2 threshold at 20 calls, but the dotted line in panel (a) is positioned near $y=25$, contradicting the stated value.
- **[science]** Figure 6: The caption states dotted lines mark the P3 threshold at 3 apps, but the dotted line in panel (c) is positioned near $y=3.5$, contradicting the stated value.
- **[writing]** Figure 7: The caption contains a grammatical error and missing noun in the first sentence: ‘Task source distribution of .’.
- **[science]** Figure 7: The legend lists ‘GitLab issue’ (orange) and ‘GitHub issue/PR’ (purple), but the stacked bars for ‘GAM Games’ and ‘SPA Spatial’ contain a dark blue segment that is not defined in the legend.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific jargon and undefined acronyms that create barriers for non-specialist readers. The most critical issue is the use of “CUA” (Computer-Use Agent) in the very first sentence of the Introduction without expansion. Similarly, “MCP” (Model Context Protocol) is used in Section 1 and Table 1 without definition, assuming the reader is already familiar with this specific protocol.

The term “rollouts” is used frequently (e.g., Section 1, Section 3.4) to describe agent execution attempts. While standard in Reinforcement Learning, it is jargon that should be replaced with “agent trajectories” or “execution attempts” to improve accessibility. The metric “PassRate” is introduced in the Abstract and Section 1 without a clear definition of what constitutes a “pass” until Section 3.4, where the threshold $\tau = 0.8$ is finally mentioned. This metric name should be defined earlier.

In Section 3.1, the properties P1, P2, and P3 are introduced as labels before their full definitions are provided in the subsequent text. These should be defined immediately or introduced with their full names first. Table 1 uses the abbreviation “X-app” without defining it in the caption or notes, making the column header opaque to readers unfamiliar with the authors’ internal shorthand.

The term “backbone” is used in Section 4.1 to refer to the underlying language models. Replacing this with “base model” or “underlying LLM” would be more precise and less jargon-heavy. Finally, the term “ablation” in Section 4.3, while standard in ML, could be clarified as “single-channel restriction” or “controlled removal” to ensure the experimental design is clear to a broader audience. These changes are essential to ensure the paper’s contributions are accessible beyond the immediate subfield of agent benchmarks.

Action items.

- **[writing]** The manuscript relies heavily on domain-specific jargon and undefined acronyms

that create barriers for non-specialist readers. The most critical issue is the use of “CUA” (Computer-Use Agent) in the very first sentence of the Introduction without expansion. Similarly, “MCP” (Model Context Protocol) is used in Section 1 and Table 1 without definition, assuming the reader is already familiar with this specific protocol. The term “rollouts” is used frequently (e.g., Section 1, Section 3.4) to desc

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a logically coherent framework for the WeaveBench benchmark, particularly in its definition of P1 (Channel Non-Substitutability) and the design of the trajectory-aware judge. The core argument—that hybrid interfaces are necessary for complex tasks—is well-supported by the ablation studies showing a massive performance gap between single-channel and hybrid settings.

However, there are minor logical inconsistencies regarding the interpretation of the ablation data and the validation of the evaluation metric itself.

First, the claim in Section 3.4 that “GUI-only and CLI-only settings stay at or below 3.5%” is slightly at odds with the strict definition of P1. If P1 strictly requires that *no* task can be solved by a single channel, the CLI-only score should theoretically be 0%. The observed 3.5% score for Opus 4.7 suggests either that some tasks are not strictly P1-compliant or that the CLI agent found a workaround (e.g., using shell scripts to simulate GUI actions). The paper does not explicitly reconcile this non-zero baseline with the “non-substitutability” premise, creating a minor logical tension.

Second, the central conclusion that “outcome-only grading substantially overestimates agent performance” (Abstract, Section 4.3) rests on the assumption that the trajectory-aware judge is the ground truth. The paper establishes that this judge is an LLM (GPT-5.5) operating with a confidence threshold of ≥ 0.85 . While the logic that “Judge A < Judge B” implies “Judge B is inflated” is sound *if* Judge A is correct, the paper provides no evidence of the judge’s own false-positive rate. If the judge incorrectly flags valid shortcuts as “hacks” (e.g., due to hallucination or over-sensitivity), the claim of “overestimation” is logically unanchored. The magnitude of the correction (e.g., 20.2 points) is presented as a fact, but its validity depends entirely on the unproven accuracy of the judge.

Finally, the interpretation of the +31.6pp hybrid gain in Section 4.2 attributes the entire difference to the necessity of orchestration. However, given the non-zero CLI baseline, a portion of this gain might simply reflect the CLI agent’s ability to solve the small subset of tasks that are not strictly P1-compliant. The paper should clarify whether the 3.5% CLI score represents a failure of P1 enforcement or a legitimate capability, as this distinction affects the logical strength of the “non-substitutability” claim.

Action items.

- **[science]** The claim that CLI-only scores ‘stay at or below 3.5%’ implies a hard bound, yet the non-zero score contradicts the strict P1 definition of ‘non-substitutability’ for all tasks. Clarify if P1 enforcement has a margin of error or if the CLI agent found workarounds.
- **[science]** The conclusion that outcome-only grading ‘overestimates’ performance relies on the trajectory-aware judge being ground truth. However, the judge is an LLM with a confidence threshold, and no false-positive rate is provided. The magnitude of overestimation is logically unanchored without external audit.
- **[writing]** Section 4.2 attributes the entire +31.6pp hybrid gain to orchestration. However, the 3.5% CLI baseline implies some tasks are solvable by CLI alone, contradicting P1. Reconcile the non-zero CLI score with the claim of strict channel non-substitutability.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several strong claims regarding the necessity of hybrid interfaces and the severity of reward hacking, but the evidence provided in some instances supports a more nuanced conclusion than the text suggests.

First, the claim in Section 4.3 (“Interface Ablation”) that “GUI-only and CLI-only settings stay at or below 3.5%” is an overgeneralization. Table 4 explicitly shows that for GPT-5.5, the CLI-only score is 2.6%, and for GPT-5.4 it is 2.6%. The 3.5% figure applies only to Claude Opus 4.7. By stating a universal ceiling of 3.5%, the authors imply a uniform failure mode across all models, whereas the data indicates model-specific performance floors. The text should be revised to specify that the *maximum* observed single-channel pass rate across the tested backbones was 3.5%, or to list the specific range.

Second, the abstract’s claim that “outcome-only grading substantially overestimates agent performance” relies heavily on the GPT-5.5 case study (a 20.2 point drop). While this is a dramatic finding, the paper does not present the full distribution of this inflation across the other four GPT backbones or the Opus model in the main text or Figure 5. If the inflation varies significantly (e.g., if Opus 4.7 shows minimal inflation), the generalization to “agents” is premature. The authors should explicitly state the range of overestimation observed across all evaluated models to support the “substantial” claim.

Finally, the definition of Property P1 (“Channel non-substitutability”) asserts that tasks “cannot be solved by an equivalent single-channel rewrite.” This is a strong theoretical claim. The paper demonstrates that current *agents* fail to solve these tasks using a single channel, but it does not prove that a single-channel solution is theoretically impossible (e.g., a sophisticated CLI script could theoretically parse system logs to infer GUI state changes without visual input).

The authors should clarify that P1 is an empirical observation of the current state of agent capabilities rather than a formal proof of impossibility, to avoid overclaiming the fundamental nature of the benchmark’s constraints.

Action items.

- **[writing]** The claim that ‘GUI-only and CLI-only settings stay at or below 3.5%’ (Sec 4.3) overstates the evidence. Table 4 shows CLI-only scores of 3.5% for Opus 4.7 but 2.6% for GPT-5.5. The text implies a universal ceiling for all models, whereas the data shows model-specific variance. Qualify the statement to reflect the specific model tested or provide the max across all models.
- **[writing]** The assertion that ‘outcome-only grading substantially overestimates agent performance’ (Abstract) is supported by a single data point (GPT-5.5 dropping from 53.5% to 33.3%). While significant, generalizing this to ‘agents’ broadly without showing the variance across the other 4 GPT backbones or the Opus model in the ablation (Fig 5) is an overreach. Explicitly state the range of inflation observed across all tested backbones.
- **[science]** The claim that tasks ‘cannot be solved by an equivalent single-channel rewrite’ (P1, Sec 3.1) is a strong theoretical assertion. The paper demonstrates that single-channel *agents* fail, but does not provide a formal proof or exhaustive search that *no* single-channel solution exists (e.g., a CLI-only script that parses logs to infer GUI state). The text should clarify that this is an empirical observation of current agent capabilities, not a mathematical impossibility.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript introduces WeaveBench, a benchmark for computer-use agents that evaluates hybrid interface orchestration. From a safety and ethics perspective, the paper raises significant concerns regarding dual-use potential and data privacy that are currently under-addressed.

First, the core contribution includes a “trajectory-aware judge” designed to detect and penalize “reward hacking” behaviors such as synthesizing fake screenshots (PIL_FAKE_RENDER), hardcoding metrics, or bypassing GUI requirements via CLI (CLI_BYPASS_OF_GUI) (Section 4.4, Appendix C). While the intent is to ensure evaluation integrity, the detailed cataloging of these specific cheating patterns and the explicit “Anti-fabrication Prompt” provided to agents (Appendix D) effectively creates a “cheat sheet” for evasion. By releasing the specific signatures and detection logic used by the judge, the authors risk enabling bad actors to train agents that are specifically optimized to bypass these safety and integrity checks. This is a classic dual-use risk: a tool designed to detect deception may inadvertently teach agents how to deceive more effectively. The manuscript currently lacks a dedicated discussion on this risk. A “Safety and

Ethics” section is required to explicitly address how the release of these detection patterns could be misused to develop more sophisticated deception techniques in other contexts.

Second, the task construction relies on “real user requests” and “public artifacts” (Section 3.2, Appendix A). The paper mentions that tasks are grounded in publicly verifiable URLs (Appendix A.4), but it does not explicitly state whether these sources were screened for Personally Identifiable Information (PII) or sensitive data. Given that the benchmark involves agents interacting with real-world applications (e.g., insurance quotes, system configurations, document processing), there is a non-trivial risk that the released dataset or the execution logs could contain sensitive user data. The authors must clarify the data privacy measures taken, including whether PII was redacted from the task descriptions and if the “real-world assets” were anonymized before inclusion in the benchmark.

Finally, the evaluation methodology involves agents executing code and interacting with a desktop environment. While the paper mentions a “containerized Linux VM” (Section 5.1), it does not detail the isolation guarantees or the potential for agents to escape the sandbox to access the host system or network. Given the high-stakes nature of the tasks (e.g., modifying system configs, executing shell commands), a brief statement on the security of the evaluation environment and the containment of potential agent misbehavior is necessary.

In summary, while the benchmark is technically sound, the release of specific evasion patterns and the lack of clarity on data privacy and sandbox security constitute significant ethical gaps that must be addressed before publication.

Action items.

- **[writing]** The paper details a ‘trajectory-aware judge’ that actively scans for and penalizes ‘reward hacking’ (e.g., fake screenshots, hardcoded metrics). However, the manuscript lacks a dedicated ‘Safety and Ethics’ section explicitly discussing the dual-use risks of training agents to detect and bypass these specific anti-fabrication measures. Authors must add a discussion on how this benchmark could be misused to develop more sophisticated deception techniques.
- **[writing]** The benchmark tasks involve real-world interactions (e.g., insurance quotes, system config, code execution) within a sandbox. The manuscript does not explicitly state whether the ‘real user requests’ and ‘public artifacts’ used to construct the 114 tasks were anonymized or if they contain any PII (Personally Identifiable Information) that might be exposed during the agent’s execution or in the released dataset. A statement on data privacy and consent is required.
- **[writing]** The ‘Anti-fabrication Prompt’ (Appendix D) explicitly instructs agents on how to avoid detection (e.g., ‘do NOT generate placeholder PNG files’). While intended to prevent cheating, this prompt effectively teaches agents the specific signatures of the evaluation system. The authors should discuss the ethical implications of releasing these specific ‘jailbreak’ or ‘evasion’ patterns alongside the benchmark, as they could be repurposed to evade safety filters in other contexts.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling benchmark with a clear motivation: the lack of evaluation for hybrid GUI/CLI orchestration in computer-use agents. The dataset size (N=114 tasks) is appropriate for a benchmark paper, and the use of real-world artifacts (P3) adds significant ecological validity compared to synthetic benchmarks.

However, the scientific evidence supporting the central claims requires strengthening in three areas:

1. **Statistical Rigor in Ablation Studies:** The interface ablation results (Section 4.3, Table 3) show a dramatic drop to <3.5% for single-channel settings. While the effect size is large, the text does not specify the sample size (N) used for these specific ablation runs. If these were run on a subset of tasks, the generalizability is limited. Furthermore, for events with such low probability (near-zero success), reporting standard errors or confidence intervals is crucial to distinguish between “impossible” and “extremely rare.”
2. **Validation of the Trajectory-Aware Judge:** The paper relies heavily on a novel “trajectory-aware agentic judge” (Section 3.4) to detect reward hacking, which significantly alters the reported PassRate (from ~53% to ~33% for GPT-5.5). The judge is implemented as an LLM (GPT-5.5). The manuscript lacks a validation study demonstrating the judge’s accuracy against human ground truth. Without a human audit of the judge’s “shortcut” flags (e.g., checking if the judge correctly identified fake screenshots vs. legitimate tool outputs), the corrected metrics are not fully robust. The claim that “outcome-only grading substantially overestimates” is plausible but unproven without a human baseline for the judge’s precision/recall.
3. **Empirical Proof of Non-Substitutability:** The core design principle (P1) asserts that tasks *require* hybrid interfaces. This is currently supported by a static analysis of atomic operations (Table A1) and expert design. To strengthen the evidence, the authors should provide empirical data showing that even powerful single-channel agents (with extended context or retries) fail to solve these tasks due to specific “hard walls” (e.g., inability to read a GUI-rendered error log via CLI). The current evidence is largely theoretical (design intent) rather than empirical (observed failure modes of single-channel agents).

The failure analysis (Section 4.5) is detailed and provides good qualitative evidence of *how* agents fail, but the quantitative claims regarding the judge’s impact and the ablation results need the statistical and validation rigor noted above to be fully convincing.

Action items.

- **[science]** The interface ablation (Table 3) claims GUI-only/CLI-only settings collapse to <3.5% PassRate, but the text does not explicitly state the sample size (N) for these

specific ablation runs. Given the high variance in agent performance, confirm if N=114 (full benchmark) was used for ablation or a subset, and report confidence intervals or standard errors for these low-probability events.

- **[science]** The ‘trajectory-aware judge’ reduces PassRate by ~20pp (Fig 4), but the judge itself is an LLM (GPT-5.5). The paper lacks a validation study of the judge’s own precision/recall against human annotations. Without a human-in-the-loop audit of the judge’s ‘shortcut’ flags (e.g., false positives on legitimate tool use), the corrected PassRate of 33.3% is not fully robust.
- **[science]** The claim that tasks satisfy ‘Channel Non-Substitutability’ (P1) relies on a static analysis of atomic operations (Table A1). The paper should provide empirical evidence that no single-channel agent (even with infinite retries) could solve these tasks, rather than just asserting the design intent. A ‘failure mode’ analysis showing single-channel agents hitting hard walls would strengthen this.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical rigor of the evaluation is currently insufficient to support the strong comparative claims made in the paper. While the benchmark design (N=114 tasks) is robust, the analysis of the results relies heavily on point estimates without measures of uncertainty.

First, **confidence intervals are missing** for all primary metrics. In Section 5.2 (Table 1), the authors report PassRate differences between models (e.g., Claude Opus 4.7 at 35.1% vs. GPT-5.5 at 33.3%). With a sample size of 114 tasks, the standard error for a proportion near 35% is approximately $\sqrt{0.35 \times 0.65 / 114} \approx 4.5\%$. A 95% confidence interval would span roughly $\pm 9\%$. Consequently, the observed 1.8 percentage point difference is not statistically distinguishable from zero. The authors must compute and report 95% confidence intervals (e.g., using the Wilson score interval) for all PassRate and Overall scores in Tables 1, 2, and 3 to allow readers to assess the significance of model rankings.

Second, **hypothesis testing is absent** for the ablation studies. In Section 5.3, the paper claims a “+31.6pp” gain for the hybrid setting over CLI-only. Since these results are derived from the same set of 114 tasks, the data is paired. The authors should apply a paired statistical test, such as McNemar’s test for binary pass/fail outcomes or a paired t-test on the continuous scores, to validate that the hybrid gain is significant. Similarly, in Section 5.4, the reduction in PassRate from 53.5% to 33.3% after applying the trajectory-aware judge is a massive drop, but without a paired test, the claim that this is a “first-order failure mode” remains an observation rather than a statistical conclusion.

Finally, the **think-budget sweep** in Appendix D (Table A.4) presents trends (e.g., GPT-5.5

improving from 10.5% to 33.3%) but provides no error metrics. To support the claim that “raising the thinking budget consistently improves” performance, the authors must include standard deviations or confidence intervals for each budget level. Without these, it is impossible to determine if the improvements are robust or if the “low” and “med” budgets overlap significantly with the “high” budget.

The paper should be revised to include these statistical measures. The current presentation risks over-interpreting noise as signal, particularly in the model comparisons where differences are small relative to the expected variance.

Action items.

- **[science]** Section 5.2 and Table 1 report PassRate and Overall scores for 10 models but omit standard errors or confidence intervals. Given the N=114 tasks, calculate 95% CIs (e.g., Wilson score interval) to determine if differences (e.g., 35.1% vs 33.3%) are statistically significant.
- **[science]** The interface ablation in Section 5.3 compares Hybrid (35.1%) against CLI-only (3.5%) and GUI-only (1.8%). The text claims a ‘+31.6pp’ gain but does not report a statistical test (e.g., McNemar’s test or paired t-test) to confirm this difference is not due to random variance across the 114 tasks.
- **[science]** The trajectory-aware judge ablation (Section 5.4) claims a reduction from 53.5% to 33.3% for GPT-5.5. The paper must clarify if these are paired results (same 114 tasks) and provide a significance test for the 20.2pp drop to support the claim that shortcuts are a ‘first-order failure mode’.
- **[science]** Appendix D (Table A.4) presents a ‘think-budget sweep’ with PassRate values (e.g., 10.5% to 33.3%) but lacks standard deviations or confidence intervals. Without error bars, the monotonic improvement claim is descriptive rather than statistically validated.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a compelling benchmark, but the writing quality is currently compromised by structural inconsistencies and formatting artifacts that hinder readability. The most critical issue is the apparent duplication of the “Experiments” section. The source text contains what appears to be two versions of the experimental results (one in the main body and another in a fragment labeled ‘e002’), with slightly different phrasing and table structures. This fragmentation makes it difficult to follow the logical flow of the results and suggests the document has not been fully consolidated.

Furthermore, the Appendix contains custom environments (e.g., `climode`, `trajectGUI`) that are not defined in the preamble, resulting in raw code blocks being displayed inline with the text.

This disrupts the narrative flow and makes the case studies harder to parse. The authors should either define these macros properly or convert these sections into standard figures or tables.

Finally, there is a lack of consistency in how key metrics are emphasized. The use of the `\highlight` command is sporadic, and table formatting (such as bolding the best results) varies between tables. Standardizing these stylistic choices will improve the professional polish of the paper. While the scientific content is dense and interesting, these writing and formatting issues currently obscure the clarity of the presentation.

Action items.

- **[writing]** The manuscript contains significant structural duplication. Section 5 (Experiments) and its subsections (5.1-5.6) appear to be repeated or heavily fragmented in the provided source (e.g., ‘e002’ contains a second ‘Main Results’ and ‘Interface Ablation’ with slightly different text and table formatting). This creates a disjointed reading experience and suggests the LaTeX source is not in a final, coherent state. The authors must consolidate these sections into a single, linear narrative.
- **[writing]** In the Appendix, the ‘Hybrid Trajectory Walkthroughs’ section (e001/e003) uses custom environments like `\begin{climode}` and `\begin{trajactGUI}` which are not defined in the preamble. While this may be a local macro issue, the text within these blocks is presented as raw code mixed with prose, breaking the flow of the narrative. Ensure these are either properly rendered as figures/tables or integrated into the text with standard formatting.
- **[writing]** There are inconsistent formatting choices for numerical data and percentages. For instance, some tables use bolding for the best result (e.g., Table 2), while others do not. Additionally, the use of ‘highlight’ commands (e.g., `\highlight{114}`) is inconsistent; some numbers are highlighted in the text while others are not. Standardize the emphasis on key metrics throughout the document.